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RV COLLEGE OF ENGINEERING®
 (An Autonomous Institution affiliated to VTU)
IV Semester B. E. Examinations Sep/Oct - 2024
Common to EI/EC/EE/ET

MICROCONTROLLER AND PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 and 9 and 10.

PART-A

M B C

1	1.1	Give an example for a processor based on typical CISC ISA.	01	2	1
	1.2	Differentiate between compiler and assembler.	01	2	1
	1.3	Which if the preferable control unit to support complex operations?	01	2	1
	1.4	What is the advantage of THUMB 2 instruction set compared to ARM ISA?	01	2	1
	1.5	Cortex M7 can deliver up to 6.5 CoreMark/mw at 1.8V. What does this statement signify?	01	2	2
	1.6	Mention the operation of debug and trace unit of Cortex M4 architecture.	01	2	1
	1.7	Identify the mode of operation to execute the following code segment: <pre>#include "main.h" int main(void){ HAL_Init(); SystemClock_Config(); MX_GPIO_Init(); While (1) { HAL_GPIO_WritePin(GPIOD,GPIO_PIN_15,GPIO_PIN_SET); HAL_Delay(1000); HAL_GPIO_WritePin(GPIOD,GPIO_PIN_15,GPIO_PIN_RESET); HAL_Delay(1000); } }</pre>	01	2	2
	1.8	Write the operation of following assembly instruction: UMULL R2,R3,R0,R1	01	2	2
	1.9	Identify the output pin configuration for STM32f407VG as shown in Fig 1.9 below.	01	2	1

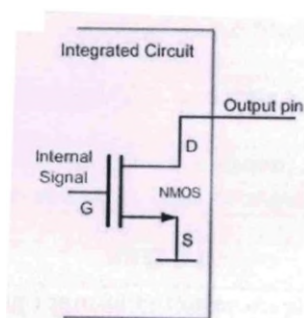
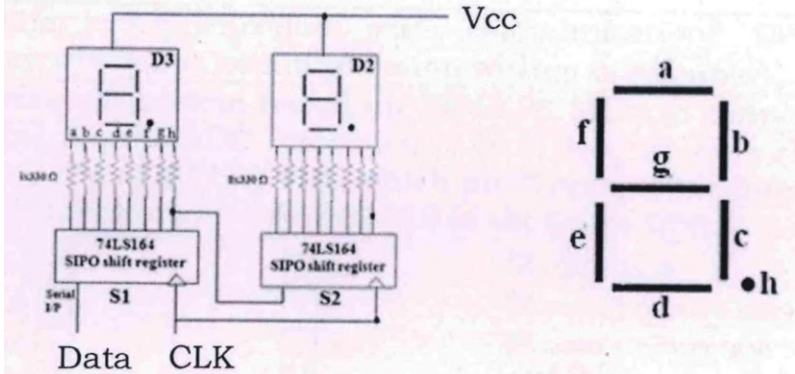
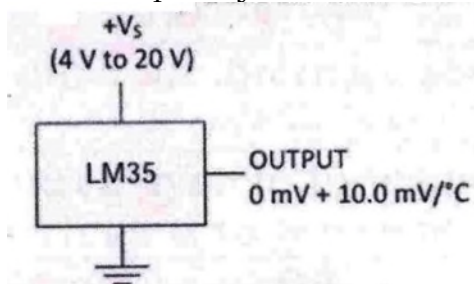


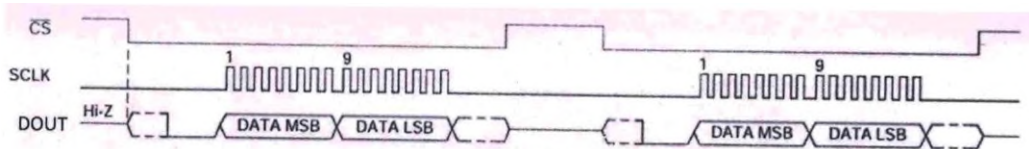
Fig 1.9

1.10	Write the C programming construct used to define the last argument of the function call as shown below. <i>HAL_GPIO_WritePin(GPIOD,GPIO_PIN_15,GPIO_PIN_SET);</i>	01	2	2
1.11	What determine the dynamic range of the input voltage for analog digital converter?	01	2	1
1.12	Name two types of DACs typically employed in digital to analog conversion.	01	1	1
1.13	What is the advantage of oversampling in serial data reception?	01	1	1
1.14	Highlight the use of NSS pin of SPI module in STM32f407VG MCU.	01	2	1
1.15	Write the value of last argument in the following API to set timeout equal to 1second. <i>HAL_StatusTypeDef HAL_SPI_Receive (SPI_HandleTypeDef *hspi, uint8_t *pData, uint16_t Size, uint32_t Timeout)</i>	01	2	2
1.16	Mention the pins used to manage hardware flow control in UART of STM32F407VG MCU.	01	1	1
1.17	Give an example for hardware interrupt in STM32F407VG MCU.	01	1	1
1.18	If the starting address of the vector table is 0x00000000 and Reset vector is located at an offset of 4bytes from starting address. Identify the location.	01	2	2
1.19	Write the size of the stack frame in Cortex M4 CPU.	01	1	1
1.20	What is the interface used in ST link debugger?	01	1	1

PART-B

2	a	What is address space? Show differences between Von Neumann and Harvard address space with suitable diagrams with examples.	06	2	2
	b	Show Single Precision and Double Precision representation of a real number 85.125	04	3	2
	c	Write a diagram showing different families of ARM processors. Highlight importance of each family with application domain.	06	2	2
3	a	Write importance of Main Stack Pointer (MSP) and Process Stack Pointer (PSP) by highlighting status of SPSEL bit of control register.	04	2	2
	b	What is the difference between LDRSB and LDRSH instructions? Explain with examples.	06	3	2
	c	Write Thumb-2 ALP to find an element in an array of elements using binary search.	06	4	3
		OR			
4	a	Differentiate between Overflow flag (V) and Sticky Overflow flag (Q) of APSR register.	04	2	2
	b	What is the difference between UMULL and UMLAL instructions? Explain with examples.	06	3	2
	c	Write Thumb-2 ALP to perform 32-bit addition, subtraction, multiplication and division on operands stored in RAM location 0 × 20000000.	06	4	3

5	a	Write a diagram showing manual and power on Reset operation. Explain the importance of watch dog and brown out reset.	06	2	2																																				
	b	Fig 5b shows an interface that provides two-digit seven segment display driven by the outputs of two cascaded serial-in-parallel-out shift registers. Data to be displayed is transmitted serially (first segment H). Each bit is clocked into the shift registers by providing common clock. Data in first shift register appears at second after eight clock pulses. Provide a push button to start the counting operation.  Fig 5b The 7-segment type is common anode type. The display codes are obtained as follows: Display codes for 3. <table border="1" data-bbox="295 861 1263 1050"><thead><tr><th>Segment</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th></tr></thead><tbody><tr><td>Bit</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Input</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Hex Code</td><td colspan="4">0</td><td colspan="4">D</td></tr></tbody></table> 7-segment display codes: 03H, 9FH, 25H, 0DH, 99H, 49H, 41H, 1FH, 01H, 09H Design STM32F407VG system interface seven-segment display and realize 2-digit BCD counter. Assume HAL is generated using STM32CubeMx tool. Write application code.	Segment	A	B	C	D	E	F	G	H	Bit	7	6	5	4	3	2	1	0	Input	0	0	0	0	1	1	0	1	Hex Code	0				D				10	4	4
Segment	A	B	C	D	E	F	G	H																																	
Bit	7	6	5	4	3	2	1	0																																	
Input	0	0	0	0	1	1	0	1																																	
Hex Code	0				D																																				
		OR																																							
6	a	Write a diagram showing basic circuit of GPIO pin. Explain briefly.	06	2	2																																				
	b	Design a suitable scheme to interface analog temperature sensor LMT86/LM35 to STM32F407VG MCU and program ADC to convert temperature value to digital value. Store the digital value in variable “TemperatureDigital”. The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly proportional to the Centigrade temperature. The pin layout is shown in Fig 6b.  Assume HAL of ADC in STM32F407VG is generated using STM32CubeMx tool. Write an application code using APIs of HAL.	10	4	4																																				

7	a	What is asynchronous serial communication? Describe framing in asynchronous communication with diagram and examples.	06	2	2
	b	Design a system based on STM32F407VG MCU to connect a sensor that requires ADC resolution of 16bits. The designers are suggested to use TI's ADS1118 ADC which provide SPI compatible interface. The pin functionality of ADS1118 is shown in Fig 7b (i).  SCLK: Serial Clock CS: Chip select (active low) NC: No Connection AIN: Analog Input DIN: Serial Data Input DOUT: Serial Data Output Figure 7(bi) Fig 7b (ii) shows the timing diagram to read16-bit digital value from ADC.  Figure 7(bii) Assume HAL of SPI in STM32F407VG is generated using STM32CubeMx tool. Write an application code using APIs of HAL to read digital value from ADC and store in a variable “ADCValue”. OR <td>10</td> <td>4</td> <td>4</td>	10	4	4
8	a	Write a diagram showing connection of a master and three slaves using SPI. Explain the data exchange operation.	06	2	2
	b	Write an application program to send “RVCE is my college” on Tx line of USART 2 at baud rate of 9600,8 data bits, 1 stop bit. Show the value loaded to USART_BRR register to achieve given baud rate at APB clock rate of 30MHz. Assume HAL is generated using STM32CubeMx tool.	10	3	2
9	a	With a diagram, explain different interrupt sources in ARM Cortex M4 core.	08	2	2
	b	Design STM32F407VG based system to toggle LED connected to PA5 at the rate of 10Hz. Use Timer 2 to generate suitable delay.	08	4	3
		OR			
10	a	What are pulse width modulators? Explain PWM module in STM32F407VG.	08	3	2
	b	Write application code in Keil IDE to use ADC to analog to digital conversion triggered by timer 2 at 100ms. The ADC output value must be validated in debug window and interrupt rate in CRO. Channel: IN1 (PA1), PCLK2:16MHz, ADCCLK: 8MHz Sampling time: 28 ADCCLKs, Resolution: 8bits Triggering (dynamic range): 0V to 3.3V (DC). Enable interrupts for ADC on conversion completion to read digital value and toggle LED on PD15 in interrupt service routine.	08	4	3